

Clipnote: AI-Powered Lecture Note-Taking App

■ LIVE SECURE APPLICATION URL (HTTPS):
<https://palace-subjective-except-stats.trycloudflare.com/>
(Direct AWS EC2 IP: <http://3.27.186.233>)

Application Name:	Clipnote	Live HTTPS URL:	https://palace-subjective-except-stats.trycloudflare.com/
Primary LLM API:	Google Gemini 2.5 Flash API	Cloud Infrastructure:	AWS EC2 (Ubuntu 24.04 / t3.micro)
Target Audience:	Students, Researchers, Educators	Tech Stack:	Next.js 16, FastAPI, SQLite, Nginx

1. Problem Statement & Objective

Modern students and self-learners consume hours of dense audio/video lectures, webinars, and recorded academic sessions daily. Manual note-taking during long lectures is inefficient, prone to missing key details, and creates a heavy cognitive load. Existing tools either offer simple verbatim transcripts without structural hierarchy or lack multi-format exports.

Core Objective: Clipnote solves this by providing an automated end-to-end pipeline that transforms raw audio/video uploads or YouTube links into structured, high-retention study packages—complete with sectioned notes, key term definitions, self-assessment quizzes, and executive study guides in seconds.

2. Target User & Use Cases

- **University & High School Students:** Converting recorded university lectures into organized revision notes and flashcards before exams.
- **Online Learners & Self-Taught Developers:** Summarizing long YouTube tutorials, coding bootcamps, and technical webinars.
- **Researchers & Educators:** Extracting core concepts and generating quick assessment quizzes from academic audio recordings.

3. LLM Models & API Architecture

Clipnote leverages **Google Gemini 2.5 Flash** (`gemini-2.5-flash`) via the google-genai SDK as its primary LLM engine. Gemini 2.5 Flash was chosen for its ultra-fast inference speed, 1M token context window, and exceptional structured JSON generation capabilities. The system features a multi-tiered API architecture:

- **Primary Processing Engine:** Google Gemini 2.5 Flash API for multi-segment transcription synthesis and structured JSON parsing.
- **Secondary Fallback Engine:** OpenAI API (`gpt-4o-mini`) automatically engages if Gemini API limits are reached.

- **Transcription Pipeline:** Multi-tiered pipeline utilizing native YouTube captions (`youtube-transcript-api`), Gemini Audio API, and OpenAI Whisper.

4. Key Application Features

Feature Module	Technical & Functional Description
Multi-Source Ingestion	Supports direct local media uploads (.mp3, .mp4, .wav, .m4a) and YouTube lecture URLs.
Interactive Media Player	Synchronized audio/video player with click-to-seek timestamp navigation across transcript segments.
Structured Note Generation	Generates hierarchical lecture notes categorized into overview, core concepts, and key takeaways.
Interactive Quiz Engine	Generates 5-question multiple choice quizzes with instant feedback, scoring, and explanation hints.
Executive Study Guide	Produces a high-yield revision checklist and key terms glossary for rapid pre-exam review.
Multi-Format Exporting	One-click exports to publication-ready PDF reports, raw Markdown (.md), and Anki Flashcard decks (.txt).

5. Expected User Experience & Outcomes

- **Time Reduction:** Reduces post-lecture review time by up to 80% by automating initial note drafting and concept organization.
- **Active Recall Learning:** Enhances long-term knowledge retention through immediate interactive quiz testing and flashcard export.
- **Seamless UI/UX:** Provides a modern dark-mode responsive web app with zero-latency tab switching and real-time task status polling.
- **Production-Ready Cloud Hosting:** Publicly hosted live on AWS EC2 (`http://3.27.186.233`) behind an Nginx reverse proxy managed by systemd and PM2.